

NAGATA COMPACTIFICATION FOR POINTED MONOID SCHEMES

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ABSTRACT. We prove that a separated finite-type morphism of pointed monoid schemes over a quasi-compact and quasi-separated base admits a proper compactification. Properness is understood in terms of congruence spaces.

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1. INTRODUCTION

The compactification problem asks whether a space over a base can be realized as an open subspace of one that is proper over the same base. For schemes admitting an immersion into projective space, taking the closure gives such a construction. Nagata's theorem shows that the existence of a compactification does not depend on the existence of a projective embedding: every separated morphism of finite type over a quasi-compact and quasi-separated scheme factors as an open immersion followed by a proper morphism. The classical theorem is due to Nagata [Nag62, Nag63]; Deligne's extension to quasi-compact and quasi-separated bases is presented by Conrad [Con07, Con09]. The aim of this paper is to establish an analogous existence theorem for pointed monoid schemes.

Monoid schemes arise in geometry over $\mathbb{F}_1$ by replacing the rings in the construction of a scheme with commutative monoids, as in Deitmar's approach [Dei05]. Here we use the pointed version of this construction developed by Cortiñas, Haesemeyer, Walker, and Weibel [CHWW15], in which the monoids have an absorbing zero and their homomorphisms preserve it. In this setting, even the formulation of properness requires some care. Quotients of monoids are defined by congruences, which may identify distinct nonzero elements, and the resulting closed immersions need not have closed image in the ordinary space of prime ideals. The congruence spaces introduced by Lorscheid and Ray [LR24]

provide a topology that records these identifications. We use their notion of properness: a morphism is proper if it is separated, of finite type, and universally closed on congruence spaces.

With this notion of properness, the compactification problem asks whether a pointed monoid scheme X over S can be embedded as an open subscheme of a pointed monoid scheme proper over S . More precisely, a *compactification* of $X \rightarrow S$ is a factorization

$$X \xrightarrow{j} \overline{X} \xrightarrow{p} S$$

with j a quasi-compact open immersion and p proper. By [Proposition 4.1](#), if X has a compactification over S , then $X \rightarrow S$ is separated and of finite type. Our main result proves the converse over a quasi-compact and quasi-separated base:

Theorem 1.1. Let S be a quasi-compact and quasi-separated pointed monoid scheme. Let $X \rightarrow S$ be a separated morphism of finite type. Then X has a compactification over S .

The more precise statement in [Theorem 4.17](#) also controls the boundary, in that the compactification may be chosen so that every point of $\overline{X}$ has an affine neighborhood $\mathrm{Spec}_0(A)$ on which

$$X \cap \mathrm{Spec}_0(A) = D(t)$$

for an element $t \in A$ whose multiplication map $A \rightarrow A$ is injective. In particular, X is schematically dense in $\overline{X}$. This local cancellation condition is obtained by blowing up a locally finitely generated ideal defining the complement of X .

The proof follows the modification and gluing approach to classical Nagata compactification presented in [[The26](#), Tag 0F3T]. For an affine scheme of finite type over an affine open of the base, an embedding in affine space gives a compactification by taking its schematic image in relative projective space. The central step is to make these affine constructions compatible. We prove that a separated morphism of finite type which is an isomorphism over a prescribed quasi-compact open becomes an open immersion after taking a strict transform along finitely many blowups of the target. The blowup ideals restrict to the unit ideal on the prescribed open. The proof of this step uses explicit fraction charts and extension of finitely generated ideals over quasi-compact and quasi-separated monoid schemes. It then permits the comparison of compactifications along overlaps and reduces the theorem to a binary gluing argument and finite induction.

Two features of monoid schemes require additional arguments in this construction. First, a proper locally closed immersion need not be a closed immersion, so separatedness of the glued scheme is verified using the image of its diagonal on congruence spaces. Second, ordinary topological density of an open does not suffice to test properness using only valuation diagrams whose generic point lies in that open; [Example 4.10](#) gives an example. The local cancellation condition on the boundary resolves this problem. After realization over a field, it makes the chosen open schematically dense in the resulting ordinary scheme. The ordinary dense valuative criterion, together with the comparison theorem of [[CHWW15](#), Theorem 8.9], then yields the restricted valuative criterion needed to finish the proof. The compactification itself and all intermediate modifications are constructed in the category of pointed monoid schemes.

Organization. [Section 2](#) recalls the monoid constructions used in the proof. [Section 3](#) fixes the notions of separatedness and properness and their valuative criteria. [Section 4](#) carries out the modification and gluing argument and proves the theorem.

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2. POINTED MONOIDS AND THEIR SCHEMES

2.1. Pointed monoids. We recall the conventions of [CHWW15, Sections 1-2].

Definition 2.1. A *pointed commutative monoid* A is a commutative monoid with identity 1 and an absorbing element 0, so that $0a = 0$ for every $a \in A$. We allow the zero monoid, in which $0 = 1$. Homomorphisms preserve 0 and 1. These objects and their homomorphisms form $\mathbf{CMon}_0$.

Its initial object is $\mathbb{F}_1 = \{0, 1\}$, and its terminal object is the zero monoid.

Definition 2.2. For $A \in \mathbf{CMon}_0$, an A -*algebra* is an object of $\mathbf{CMon}_{0,A/}$. The polynomial monoid $A[T_1, \dots, T_n]$ is the free A -algebra on n generators, in that for $B \in \mathbf{CMon}_{0,A/}$, evaluation gives a natural bijection

$$\mathrm{Hom}_{\mathbf{CMon}_{0,A/}}(A[T_1, \dots, T_n], B) \simeq B^n.$$

Explicitly, it consists of zero and the formal monomials $aT_1^{e_1} \cdots T_n^{e_n}$ with $a \neq 0$ and $e_i \geq 0$.

Definition 2.3. A morphism $A \rightarrow B$ in $\mathbf{CMon}_0$ is *of finite type* if finitely many elements generate B over the image of A , or equivalently if it admits a surjection $A[T_1, \dots, T_n] \rightarrow B$.

Definition 2.4. Let $A \in \mathbf{CMon}_0$ be a pointed commutative monoid. An *ideal* is a subset $I \subseteq A$ containing zero and satisfying $AI \subseteq I$, or equivalently a pointed A -subset of A . It is *prime* if it is proper and its complement is multiplicatively closed.

Thus $(a_1, \dots, a_r) = \bigcup_i Aa_i$, and the union of two ideals is again an ideal. Every nonzero monoid has a unique maximal ideal, namely $A \setminus A^\times$.

Definition 2.5. Let $A \in \mathbf{CMon}_0$ be a pointed commutative monoid. A *congruence* $\mathfrak{c}$ on A is an equivalence relation compatible with multiplication. The *quotient* $A/\mathfrak{c} \in \mathbf{CMon}_{0,A/}$ is initial among A -algebras whose structure morphisms identify every pair in $\mathfrak{c}$. The *null ideal* of $\mathfrak{c}$ is $\{a \mid a \sim_{\mathfrak{c}} 0\}$.

Every surjective morphism in $\mathbf{CMon}_0$ is the quotient by its kernel congruence. An ideal determines the congruence which identifies its elements with zero and makes no further identifications; general congruences also identify nonzero elements. We write A/I for the quotient attached to an ideal in this way.

Definition 2.6. Let $A \in \mathbf{CMon}_0$ be a pointed commutative monoid and let $T \subseteq A$ be a multiplicatively closed subset containing 1. The localization $T^{-1}A$ is initial among $B \in \mathbf{CMon}_{0,A/}$ in which the image of every element of T is invertible.

Explicitly, its elements are fractions a/t , with equality given by [CHWW15, Section 1]

$$a/t = b/u \iff vua = vtb \text{ for some } v \in T.$$

We write $A[t^{-1}]$ when T consists of the powers of one element.

Definition 2.7. Let $A \in \mathbf{CMon}_0$ be a pointed commutative monoid. An element $t \in A$ is *cancellable* if $ta = tb$ implies $a = b$ for all $a, b \in A$, including when one of these elements is zero. The monoid A is *cancellative* if it is nonzero and every nonzero element is cancellable. A nonzero object $K \in \mathbf{CMon}_0$ is a *pointed group* if every nonzero element is invertible. For cancellative A , its *pointed fraction group* is the group completion of $A \setminus \{0\}$ with an absorbing zero adjoined, regarded as an object of $\mathbf{CMon}_{0,A/}$.

Remark 2.8. Let $A \in \mathbf{CMon}_0$ be a pointed commutative monoid. An element $t \in A$ is cancellable if and only if $A \rightarrow A[t^{-1}]$ is injective. For cancellative A , the morphism to its pointed fraction group is injective.

Notation 2.9. For morphisms $C \rightarrow A$ and $C \rightarrow B$ in $\mathbf{CMon}_0$, write $A \amalg_C B$ for their pushout. Thus, for $D \in \mathbf{CMon}_0$, restriction induces a natural bijection

$$\mathrm{Hom}_{\mathbf{CMon}_0}(A \amalg_C B, D) \simeq \mathrm{Hom}_{\mathbf{CMon}_0}(A, D) \times_{\mathrm{Hom}_{\mathbf{CMon}_0}(C, D)} \mathrm{Hom}_{\mathbf{CMon}_0}(B, D).$$

It is constructed by generators from A and B , subject to their multiplication relations and the identifications induced by C .

2.2. Monoid schemes. The construction by spectra and gluing follows Deitmar's monoid-scheme approach [Dei05, Sections 1-3], with the pointed conventions of [CHWW15, Sections 1-2].

Definition 2.10. Let $A \in \mathbf{CMon}_0$ be a pointed commutative monoid. The *spectrum* $\mathrm{Spec}_0(A)$ consists of the prime ideals of A , with principal open subsets $D(a)$ and closed subsets $V(I)$ given by

$$D(a) = \{\mathfrak{p} \mid a \notin \mathfrak{p}\}, \quad V(I) = \{\mathfrak{p} \mid I \subseteq \mathfrak{p}\}.$$

Its structure sheaf is characterized by $\mathcal{O}(D(a)) = A[a^{-1}]$. A *pointed monoid scheme* is a space with a sheaf of pointed monoids locally isomorphic to one of these spectra. It is *affine* if it is isomorphic to $\mathrm{Spec}_0(B)$ for some $B \in \mathbf{CMon}_0$. A morphism consists of a continuous map $f : X \rightarrow Y$ and a morphism $f^\# : f^{-1}\mathcal{O}_Y \rightarrow \mathcal{O}_X$ of sheaves of pointed monoids inducing local homomorphisms on stalks. A homomorphism in $\mathbf{CMon}_0$ is *local* when the inverse image of the units is the group of units. These schemes and their morphisms form the category Sch_0 .

The spectrum construction defines a fully faithful functor

$$\mathrm{Spec}_0 : (\mathbf{CMon}_0)^{\mathrm{op}} \rightarrow \mathrm{Sch}_0$$

whose essential image consists of the affine schemes; see [CHWW15, Section 2] and, for the unpointed formulation, [Dei05, Proposition 2.2(b)]. In particular, for $A, B \in \mathbf{CMon}_0$ there is a natural bijection

$$\mathrm{Hom}_{\mathrm{Sch}_0}(\mathrm{Spec}_0(B), \mathrm{Spec}_0(A)) \simeq \mathrm{Hom}_{\mathbf{CMon}_0}(A, B).$$

Global sections recover the coordinate monoid of an affine scheme. The zero monoid corresponds to the empty scheme.

Definition 2.11. Let $f : X \rightarrow Y$ be a morphism in Sch_0 . It is an *open immersion* if it identifies X with an open subset $U \subseteq Y$ equipped with the restricted structure sheaf $\mathcal{O}_Y|_U$. It is *affine* if the inverse image of every affine open subscheme of Y is affine. It is a *closed immersion* if it is affine and its coordinate homomorphisms in $\mathbf{CMon}_0$ are surjective; see [CHWW15, Definition 2.5 and Lemma 2.7].

Every point of a nonempty affine monoid scheme is a generalization of its unique closed point. In particular, every open neighborhood of that closed point is the whole affine scheme. This gives the following observation; compare [CHWW15, Lemmas 1.3 and 2.4].

Lemma 2.12. Let $A \in \mathbf{CMon}_0$ be a pointed commutative monoid, and let $W \in \mathrm{Sch}_{0,/\mathrm{Spec}_0(A)}$ be a nonempty affine open subscheme. Then $W = D(a)$ for some $a \in A$.

Proof. Let x be the unique closed point of the affine open W . Choose a principal open $D(a)$ with $x \in D(a) \subseteq W$. Every point of W generalizes x , so belongs to $D(a)$. $\square$

Definition 2.13. For $X \in \mathrm{Sch}_0$ a pointed monoid scheme, *ideal sheaves* are subobjects of $\mathcal{O}_X$ in the category of sheaves of pointed $\mathcal{O}_X$ -sets. Here a pointed $\mathcal{O}_X$ -set is a sheaf of pointed sets with a unital $\mathcal{O}_X$ -action for which zero acts as the constant map to the base point. An ideal sheaf is *quasi-coherent* if on every affine chart it is obtained by localizing an ideal of the coordinate monoid. It is *locally finitely generated* if those ideals are locally generated by finitely many elements. Such an ideal is *invertible* if locally it has the form $t\mathcal{O}$ with multiplication by t injective, so that $\mathcal{O} \rightarrow t\mathcal{O}$ is an isomorphism of pointed sets with monoid action.

Notation 2.14. The notation $V(\mathcal{I})$ always refers to the ordinary closed subset defined by an ideal sheaf.

Remark 2.15. Affine duality identifies pushouts in $\mathbf{CMon}_0$ with fiber products of affine objects in $\mathbf{Sch}_0$, in that

$$\mathrm{Spec}_0(A) \times_{\mathrm{Spec}_0(C)} \mathrm{Spec}_0(B) = \mathrm{Spec}_0(A \amalg_C B).$$

Gluing these affine products constructs fiber products in $\mathbf{Sch}_0$. For compatible open immersions, identifying the structure sheaves on overlaps constructs a gluing in $\mathbf{Sch}_0$; morphisms out of it are exactly compatible morphisms out of its open pieces.

Definition 2.16. Let k be a field and let $A \in \mathbf{CMon}_0$ be a pointed commutative monoid. The *monoid algebra* $k[A]$ is the free k -module on $A \setminus \{0\}$, with multiplication induced by A and the monoid zero identified with the ring zero. For a pointed monoid scheme X , gluing $\mathrm{Spec}(k[A])$ over its affine charts defines its *k -realization* X_k . This construction defines a functor from $\mathbf{Sch}_0$ to the category of k -schemes.

Proposition 2.17. Realization preserves open immersions, fiber products, morphisms of finite type, closed immersions, and separated morphisms.

Proof. Monoid algebras commute with localization and carry pushouts to tensor products:

$$k[A[t^{-1}]] \simeq k[A][t^{-1}], \quad k[A \amalg_C B] \simeq k[A] \otimes_{k[C]} k[B].$$

These identities give compatibility with open immersions and fiber products by gluing. Finite generation and surjectivity of monoid homomorphisms pass to the induced algebra homomorphisms, giving preservation of finite type morphisms and closed immersions. Compatibility with fiber products identifies the realized diagonal with the diagonal of the realized morphism, so separatedness is preserved as well. See [CHWW15, Corollary 5.4 and Propositions 5.11 and 5.13]. $\square$

Definition 2.18. Let $X \in \mathbf{Sch}_0$ be a pointed monoid scheme. It is *quasi-compact* if its underlying topological space is quasi-compact, and *quasi-separated* if intersections of affine open subschemes are quasi-compact. A morphism $f : X \rightarrow Y$ in $\mathbf{Sch}_0$ is *quasi-compact* if the inverse image of every affine open subscheme of Y is quasi-compact.

Definition 2.19. Let $h : Z \rightarrow Y$ be a quasi-compact morphism in $\mathbf{Sch}_0$ with Z, Y quasi-compact and quasi-separated. Its *schematic image* is the smallest closed subobject of Y through which h factors. Equivalently, it is initial among closed subschemes of Y equipped with a factorization of h .

The following construction of the schematic image is given in [CHWW15, Proposition 9.3].

Construction 2.20. Let $h : Z \rightarrow Y$ be as in Definition 2.19. On an affine open $V = \mathrm{Spec}_0(A) \subseteq Y$, take the quotient of A by

$$a \sim b \iff h^\#(a) = h^\#(b) \text{ in } \Gamma(h^{-1}(V), \mathcal{O}_Z).$$

These quotients glue and satisfy the stated factorization property. Indeed, sections on $h^{-1}(V)$ are computed by a finite equalizer using affine charts and principal covers of their intersections. Localization commutes with these finite equalizers and with images, so the construction commutes with principal localization and hence with open restriction.

Definition 2.21. An open immersion $j : U \hookrightarrow X$ in $\mathbf{Sch}_0$ is *schematically dense* if $\mathcal{O}_X \rightarrow j_*\mathcal{O}_U$ is injective.

Remark 2.22. If every point of X has an affine neighborhood $V = \mathrm{Spec}_0(A)$ on which $U \cap V = D(t)$ with t cancellable, then $U \hookrightarrow X$ is schematically dense, since $A \rightarrow A[t^{-1}]$ is injective.

Definition 2.23. Let $g : Z \rightarrow Y$ and $p : Y' \rightarrow Y$ be morphisms in Sch_0 between quasi-compact and quasi-separated schemes, and let $U \in \text{Sch}_{0,/Y}$ be a quasi-compact open subscheme over which both morphisms are isomorphisms. The *strict transform* of g along p , relative to U , is the schematic image of the induced morphism $U \rightarrow Z \times_Y Y'$, regarded as an object of $\text{Sch}_{0,/Y'}$.

Thus strict transform is obtained by base change followed by schematic image.

Lemma 2.24. Let $Z \rightarrow Y$ and $Y_2 \rightarrow Y_1 \rightarrow Y$ be morphisms between quasi-compact and quasi-separated monoid schemes, all of which are isomorphisms over a quasi-compact open $U \subseteq Y$. Let Z_1 be the strict transform of $Z \rightarrow Y$ along $Y_1 \rightarrow Y$, relative to U . Then the schematic image of U in $Z_1 \times_{Y_1} Y_2$ is canonically the strict transform of $Z \rightarrow Y$ along the composite $Y_2 \rightarrow Y$.

Moreover, if $h : E \rightarrow P$ is a quasi-compact morphism between quasi-compact and quasi-separated monoid schemes and $V \subseteq E$ is a quasi-compact schematically dense open, then h and $h|_V$ have the same schematic image in P .

Proof. The morphism $Z_1 \times_{Y_1} Y_2 \rightarrow Z \times_Y Y_2$ is a closed immersion, and the map from U factors through it. The smallest closed subscheme of $Z \times_Y Y_2$ through which this map factors is therefore contained in $Z_1 \times_{Y_1} Y_2$. Since closed immersions compose, its minimality characterizes it equally as the schematic image in that closed subscheme.

For the second assertion, schematic density gives an injection

$$\Gamma(h^{-1}(Q), \mathcal{O}_E) \hookrightarrow \Gamma(h^{-1}(Q) \cap V, \mathcal{O}_V)$$

for every affine open $Q \subseteq P$. The two restriction maps from $\mathcal{O}_P(Q)$ thus have the same kernel congruence, which proves the assertion. $\square$

We recall the projective construction of [CHWW15, Section 7].

Definition 2.25. Let $R \in \text{CMon}_0$ be equipped with a nonnegative grading $R = \bigvee_{d \geq 0} R_d$. Its *projective spectrum* $\text{MProj}(R) \in \text{Sch}_{0,/\text{Spec}_0(R_0)}$ is obtained by gluing the affine schemes with coordinates $(R[f^{-1}])_0$ for positive-degree homogeneous f along their canonical localization overlaps. For $A \in \text{CMon}_0$ and $n \geq 0$, put $\mathbb{P}_A^n = \text{MProj}(A[T_0, \dots, T_n])$, with each T_i of degree one.

The i th chart of $\mathbb{P}_A^n$ has coordinates

$$A[T_0/T_i, \dots, T_n/T_i], \quad T_i/T_i = 1.$$

The overlap of charts i and j is obtained by inverting T_j/T_i . These constructions commute with localization in A and thus define $\mathbb{P}_S^n$ for every base S . A graded surjection from $A[T_0, \dots, T_n]$ induces a closed immersion on the resulting projective schemes [CHWW15, Lemma 7.2].

The following blowup construction is given in [CHWW15, Definition 7.4 and the preceding discussion].

Definition 2.26. Let $Y \in \text{Sch}_0$ be a pointed monoid scheme and let $\mathcal{I} \subseteq \mathcal{O}_Y$ be a quasi-coherent locally finitely generated ideal sheaf of pointed $\mathcal{O}_Y$ -sets. The *blowup* $\text{Bl}_{\mathcal{I}}(Y) \in \text{Sch}_{0,/Y}$ is obtained by gluing the projective spectra of the Rees monoids

$$\bigvee_{d \geq 0} I^d z^d, \quad I^0 = A,$$

over affine charts $\text{Spec}_0(A) \subseteq Y$, where $A \in \text{CMon}_0$ and $I \subseteq A$ represents $\mathcal{I}$.

We will prove the needed properties of its fraction charts in Section 4.

3. FINITE TYPE, SEPARATED AND PROPER MORPHISMS

3.1. Congruence spaces and properness. Let $A \in \mathbf{CMon}_0$ be a pointed commutative monoid. A closed immersion is locally of the form $\mathrm{Spec}_0(A/\mathfrak{c}) \rightarrow \mathrm{Spec}_0(A)$ for a congruence $\mathfrak{c}$. Its image need not be closed in the ordinary space of prime ideals. For example, $\mathbb{F}_1[T] \rightarrow \mathbb{F}_1$, $T \mapsto 1$, defines a closed immersion whose image is the nonclosed point (0) of the affine line. This motivates the use of congruence spaces to study separatedness and properness.

Definition 3.1. Let $A \in \mathbf{CMon}_0$ be a pointed commutative monoid. A *prime congruence* of A is a proper congruence $\mathfrak{p}$ such that $A/\mathfrak{p}$ is cancellative. Write $\mathrm{Cong}(A)$ for their set, with topology generated by

$$U_{a,b} = \{\mathfrak{p} \mid a \not\sim_{\mathfrak{p}} b\}, \quad a, b \in A.$$

These sets form a subbasis (and not necessarily a basis). A homomorphism induces a continuous map in the opposite direction by pulling back congruences. Localization at t identifies $\mathrm{Cong}(A[t^{-1}])$ with $U_{t,0}$. Consequently these spaces glue over the affine charts of a scheme X to a space $\mathrm{Cong}(X)$. These constructions are due to Lorscheid and Ray [LR24, Propositions 2.8 and 2.13].

Definition 3.2. Let $X \in \mathrm{Sch}_0$ be a pointed monoid scheme. The *null-ideal map* $\pi_X : \mathrm{Cong}(X) \rightarrow X$ is defined on each affine chart $\mathrm{Spec}_0(A) \subseteq X$ by

$$\mathfrak{p} \mapsto \{a \in A \mid a \sim_{\mathfrak{p}} 0\}.$$

This is a prime ideal because $A/\mathfrak{p}$ is cancellative, and these maps agree on overlaps.

The map π_X is continuous and surjective [LR24, Proposition 2.14]. Surjectivity can be checked by sending the complement of a prime ideal to 1 and the prime ideal to 0 in $\mathbb{F}_1$. For an open $U \subseteq X$, its inverse image is $\mathrm{Cong}(U)$. On an affine chart, $\mathfrak{p}$ specializes to $\mathfrak{q}$ precisely when $\mathfrak{p} \subseteq \mathfrak{q}$. A closed immersion defined by $\mathfrak{c}$ induces the closed embedding onto $\{\mathfrak{p} \mid \mathfrak{c} \subseteq \mathfrak{p}\}$ in the congruence space.

Definition 3.3. Let $f : X \rightarrow S$ be a morphism in Sch_0 . It is *locally of finite type* if it is locally given by a finitely generated homomorphism of monoids. It is *of finite type* if it is also quasi-compact. It is *quasi-separated* if its diagonal is quasi-compact, and *separated* if its diagonal is a closed immersion.

A separated morphism is therefore quasi-separated.

Definition 3.4. Let $f : X \rightarrow S$ be a morphism in Sch_0 . We call f *universally closed* if for every $S' \in \mathrm{Sch}_{0,/S}$ the map

$$\mathrm{Cong}(X \times_S S') \rightarrow \mathrm{Cong}(S')$$

is closed. The morphism f is *proper* if it is separated, of finite type, and universally closed.

These are the definitions of [LR24, Definitions 6.2, 7.1, and 8.1].

3.2. The valuative criterion.

Definition 3.5. Let $K \in \mathbf{CMon}_0$ be a pointed group and let $R \subseteq K$ be a pointed submonoid. The monoid $R \in \mathbf{CMon}_0$ is a *valuation monoid* with pointed fraction group K if each $x \in K \setminus \{0\}$ satisfies $x \in R$ or $x^{-1} \in R$.

For a morphism $f : X \rightarrow S$ in Sch_0 , consider diagrams

$$\begin{array}{ccc} \mathrm{Spec}_0(K) & \longrightarrow & X \\ \downarrow & \nearrow & \downarrow f \\ \mathrm{Spec}_0(R) & \longrightarrow & S \end{array}$$

Here $R \in \mathbf{CMon}_0$ is a valuation monoid and $K \in \mathbf{CMon}_{0,R/}$ is its pointed fraction group. The top arrow is arbitrary, including maps which send nonzero sections on X to zero. The following are the valuative criteria of [LR24, Theorems 6.6, 7.7, and 8.2].

Theorem 3.6. Let $f : X \rightarrow S$ be a morphism in Sch_0 . In the diagrams above, let $R \in \text{CMon}_0$ range over valuation monoids and let $K \in \text{CMon}_{0,R}/$ be its pointed fraction group.

- (1) If f is quasi-compact, it is universally closed if and only if every such diagram admits a lift.
- (2) If f is quasi-separated, it is separated if and only if every such diagram admits at most one lift.
- (3) If f is of finite type and quasi-separated, it is proper if and only if every such diagram admits a unique lift.

The last formulation retains the quasi-separatedness hypothesis needed when invoking the uniqueness criterion. All applications below satisfy it. The unique closed point of $\text{Spec}_0(R)$ also makes these tests local on the target: every affine neighborhood of its image contains the image of the whole valuation spectrum.

The following comparison is [CHWW15, Theorem 8.9].

Theorem 3.7. Let $f : X \rightarrow S$ be a morphism in Sch_0 and let k be a field. Every valuation-monoid diagram for f admits a unique lift (respectively, at most one lift) if and only if every valuation-ring diagram for $f_k : X_k \rightarrow S_k$ admits a unique lift (respectively, at most one lift). Here the ordinary diagrams use k -algebra valuation rings and their fraction fields, and all arrows are morphisms of k -schemes.

Proposition 3.8. In Sch_0 , morphisms of finite type, separated morphisms, and proper morphisms are stable under composition and base change, and each property is local on the target. Closed immersions in Sch_0 are proper.

Proof. For finite type morphisms, composition and base change reduce to finite generation in CMon_0 , together with the corresponding properties of quasi-compact morphisms. Separatedness is stable under base change and composition by the corresponding properties of closed immersions and the factorization of a composite diagonal. Both properties are local on the target. The assertions about proper morphisms and closed immersions are [LR24, Lemma 8.4]. $\square$

Remark 3.9. Let $X \rightarrow S$ be a morphism of finite type in Sch_0 and let $j : U \hookrightarrow X$ be a quasi-compact open immersion. Then $U \rightarrow S$ is of finite type. Indeed, j is locally of finite type because principal localizations are generated by the inverse of one element. Since j is quasi-compact, it is of finite type, and the assertion follows by composition.

Proposition 3.10. For $S \in \text{Sch}_0$ a pointed monoid scheme and $n \geq 0$, the structure morphism $\mathbb{P}_S^n \rightarrow S$ is proper. Consequently, a morphism in Sch_0 which locally on its target factors as a closed immersion into relative projective space followed by the structure morphism is proper.

Proof. Projective morphisms are proper by [LR24, Corollary 8.5]. The consequence follows from Proposition 3.8. $\square$

We will use the graph factorization of the following result repeatedly.

Lemma 3.11. Let $S \in \text{Sch}_0$ be a pointed monoid scheme, and let $Z, Y \in \text{Sch}_{0,S}$ have proper and separated structure morphisms, respectively. Then every S -morphism $Z \rightarrow Y$ is proper.

Proof. Factor the morphism as its graph followed by projection

$$Z \hookrightarrow Z \times_S Y \rightarrow Y.$$

The graph is a closed immersion by separatedness of $Y \rightarrow S$, and the projection is a base change of the proper morphism $Z \rightarrow S$. Both morphisms are proper by Proposition 3.8, so their composite is proper. $\square$

4. NAGATA COMPACTIFICATION

The expectation behing Nagata compactification for pointed monoid schemes comes from the:

Proposition 4.1. If X has a compactification over S , then $X \rightarrow S$ is separated and of finite type, without every quasi-compactness or quasi-separatedness hypothesis on S .

Proof. Write the compactification as $X \xrightarrow{j} \overline{X} \xrightarrow{p} S$. The open immersion j is separated, since its relative diagonal is an isomorphism. It is locally of finite type: on an affine open $\mathrm{Spec}_0(A) \subseteq \overline{X}$, the open $X \cap \mathrm{Spec}_0(A)$ is covered by principal opens $D(a)$, and $A[a^{-1}]$ is generated over A by the single element a^{-1} . Since j is quasi-compact by definition of compactification, it is of finite type. The proper morphism p is separated and of finite type by definition, so the assertion follows by stability of these properties under composition; see [Proposition 3.8](#). $\square$

We prove the theorem by constructing affine compactifications and then gluing after blowups supported on their boundaries.

Definition 4.2. Let $Y \in \mathrm{Sch}_0$ be a pointed monoid scheme and let $U \in \mathrm{Sch}_{0,/Y}$ be an open subscheme. A *modification preserving U* is a morphism $Y' \rightarrow Y$ in Sch_0 obtained as a finite tower of blowups in quasi-coherent locally finitely generated ideal sheaves on the successive schemes, each restricting to the unit ideal on the inverse image of U .

Such a morphism is proper and an isomorphism over U (but need not be surjective).

4.1. Affine compactifications and blowups.

Proposition 4.3. Let $S \in \mathrm{Sch}_0$ be a quasi-compact and quasi-separated pointed monoid scheme, let $S_0 \in \mathrm{Sch}_{0,/S}$ be an affine open subscheme, and let $X \in \mathrm{Sch}_{0,/S_0}$ be affine and of finite type. View X as an object of $\mathrm{Sch}_{0,/S}$ by composition. Then, for some $n \geq 0$, there is a closed subscheme $\overline{X} \in \mathrm{Sch}_{0,/\mathbb{P}_S^n}$ and a schematically dense open immersion $X \hookrightarrow \overline{X}$ in $\mathrm{Sch}_{0,/S}$. In particular, X has a proper compactification over S .

Proof. Choose generators of the coordinate monoid of X over that of S_0 . They give a closed immersion $X \hookrightarrow \mathbb{A}_{S_0}^n$. Let $\overline{X}$ be the schematic image of the composite $X \rightarrow \mathbb{P}_S^n$. This schematic image is defined since the composite is quasi-compact and its target is quasi-compact and quasi-separated. Its restriction to the open $\mathbb{A}_{S_0}^n$ is X , by open restriction of schematic images. Thus $X \hookrightarrow \overline{X}$ is an open immersion, schematically dense by construction, and $\overline{X} \rightarrow S$ is proper by [Proposition 3.10](#). $\square$

We characterize the blowup by its universal property in the slice over the base, and then construct it by fraction charts over arbitrary coordinate monoids.

Lemma 4.4. Let $Y \in \mathrm{Sch}_0$ be a pointed monoid scheme, and let $\mathcal{I} \subseteq \mathcal{O}_Y$ be a quasi-coherent locally finitely generated ideal sheaf of pointed $\mathcal{O}_Y$ -sets. The Rees-monoid blowup $\mathrm{Bl}_{\mathcal{I}}(Y) \in \mathrm{Sch}_{0,/Y}$ is terminal among objects $g : T \rightarrow Y$ of $\mathrm{Sch}_{0,/Y}$ for which $\mathcal{I}\mathcal{O}_T$ is invertible. Its structure morphism p is proper. It is an isomorphism over $Y \setminus V(\mathcal{I})$. On an affine chart $\mathrm{Spec}_0(A)$, with $A \in \mathrm{CMon}_0$ and $I = (a_1, \dots, a_r) \subseteq A$ representing $\mathcal{I}$, its a_i -chart has coordinate monoid $A[I/a_i] \in \mathrm{CMon}_{0,A/}$ given by

$$A[I/a_i] = \{x/a_i^n \mid x \in I^n, n \geq 0\} \subseteq A[a_i^{-1}].$$

Multiplication by a_i on this chart is injective, including on zero. If multiplication by $t \in A$ is injective on A , it remains injective on every blowup chart.

Proof. We first construct the terminal object on affine charts and verify its factorization property. The degree-zero localization of the Rees monoid $\bigvee_{n \geq 0} I^n z^n$ at $a_i z$ has the indicated fraction description. Indeed, equality of two homogeneous fractions is witnessed by multiplication by a further power of a_i , exactly as in $A[a_i^{-1}]$. The charts glue under the usual further localizations. The element a_i is

invertible in the ambient localization, so multiplication by it is injective on the chart. Every element of I is a multiple of a_i there, and the extended ideal is the principal invertible ideal (a_i) . The chart is empty if its coordinate monoid is zero.

Suppose that $A \rightarrow B$ pulls I back to an invertible ideal. Locally on $\text{Spec}_0(B)$, choose its generator b so that multiplication $B \rightarrow bB$ is an isomorphism of pointed B -sets. Write the images of the a_i as bc_i . Since b belongs to the ideal generated by these images, there are an index i and $d \in B$ with $b = bc_id$. Injectivity of multiplication by b gives $c_id = 1$. Thus the image of a_i is also a generator and is cancellable. Division by this generator defines the map from $A[I/a_i]$ to B , and cancellability verifies the fraction relations and uniqueness. These local maps agree on chart overlaps and glue to the unique lift. For the unit ideal the blowup is the identity, giving the assertion over $Y \setminus V(\mathcal{I})$.

The finitely many degree-one generators give a graded surjection from $A[T_1, \dots, T_r]$ onto the Rees monoid. Consequently the blowup is locally a closed subscheme of relative projective space. Properness follows from [Proposition 3.10](#). For the zero ideal the blowup is empty and is proper as well. Finally, an equality after localization can be cleared by a power of a_i ; cancelling t in A then proves that t remains cancellable in $A[a_i^{-1}]$ and in its chart submonoid. No finite generation of A is used. $\square$

The following is the boundary-separation step corresponding to [\[The26, Tag 0F3V\]](#). For monoid ideals, dividing by the exceptional ideal gives the required closed extensions directly.

Lemma 4.5. Let $Y \in \text{Sch}_0$ be a pointed monoid scheme, let $U \in \text{Sch}_{0/Y}$ be an open subscheme, and let $\mathcal{I}, \mathcal{J} \subseteq \mathcal{O}_Y$ be quasi-coherent locally finitely generated ideal sheaves of pointed $\mathcal{O}_Y$ -sets such that

$$V(\mathcal{I}) \cap V(\mathcal{J}) \cap U = \emptyset.$$

The blowup $Y' \in \text{Sch}_{0/Y}$ of the ideal generated by $\mathcal{I} \cup \mathcal{J}$ has proper structure morphism and is an isomorphism over U . On Y' there are disjoint closed subsets, defined by locally finitely generated quasi-coherent ideals, whose intersections with U are $V(\mathcal{I}) \cap U$ and $V(\mathcal{J}) \cap U$. In particular, the ordinary closures of these two subsets of U in Y' are disjoint.

Proof. Write $\mathcal{K} = (\mathcal{I} \cup \mathcal{J})$ and $\mathcal{E} = \mathcal{K}\mathcal{O}_{Y'}$. The hypothesis says that $\mathcal{K}|_U = \mathcal{O}_U$, so [Lemma 4.4](#) gives properness and the isomorphism over U . Divide $\mathcal{I}\mathcal{O}_{Y'}$ and $\mathcal{J}\mathcal{O}_{Y'}$ by the invertible ideal $\mathcal{E}$. On the chart associated with a generator a of $\mathcal{K}$, the resulting ideals are generated by the fractions i/a and j/a , respectively. Choose the generators a from generators of $\mathcal{I}$ and $\mathcal{J}$. On each chart one of the divided ideals contains 1, so their vanishing sets are disjoint. Over U the exceptional ideal is the unit ideal, hence these vanishing sets restrict to the prescribed subsets. As closed extensions, they contain their ordinary closures. $\square$

We first record the extension of ideals required to make local blowups global.

Lemma 4.6. Let $Y \in \text{Sch}_0$ be a quasi-compact and quasi-separated pointed monoid scheme, and let $W \in \text{Sch}_{0/Y}$ be a quasi-compact open subscheme. Let $\mathcal{K} \subseteq \mathcal{O}_Y$ be a quasi-coherent ideal sheaf of pointed $\mathcal{O}_Y$ -sets. Every quasi-coherent locally finitely generated ideal $\mathcal{J}_W \subseteq \mathcal{K}|_W$ in sheaves of pointed $\mathcal{O}_W$ -sets extends to a quasi-coherent locally finitely generated ideal $\mathcal{J} \subseteq \mathcal{K}$ in sheaves of pointed $\mathcal{O}_Y$ -sets. Moreover, there is a quasi-coherent locally finitely generated ideal sheaf $\mathcal{I} \subseteq \mathcal{O}_Y$ of pointed $\mathcal{O}_Y$ -sets with $Y \setminus V(\mathcal{I}) = W$.

Proof. Suppose first that $Y = \text{Spec}_0(A)$, and write $W = \bigcup_{i=1}^r D(s_i)$. Choose finitely many generators $a_{ij}/s_i^{n_{ij}}$ of $\mathcal{J}_W(D(s_i))$. Compatibility on $D(s_i s_k)$ implies that a sufficiently large power of s_i times a_{ij} belongs to $\mathcal{J}_W(D(s_k))$ for every k . Membership in $\mathcal{K}(A)[s_i^{-1}]$ also implies that a sufficiently large power of s_i times a_{ij} belongs to $\mathcal{K}(A)$. There are finitely many generators and overlaps, so choose such powers simultaneously. The resulting elements generate an ideal of A contained in $\mathcal{K}(A)$ and restricting to $\mathcal{J}_W$ on W .

For general Y , choose a finite affine open cover and add its members to W one at a time. At each step the intersection of the next affine with the open already treated is quasi-compact. The affine

construction extends the current ideal over that affine, and equality on the intersection permits gluing. This proves the first assertion.

For the second assertion, let $\mathcal{K}$ be the radical ideal of $Y \setminus W$. On an affine open it consists of the sections a for which $D(a) \subseteq W$. This description commutes with principal localization, so $\mathcal{K}$ is quasi-coherent. Choose finitely many principal opens in affine charts covering W and contained in W . Their defining sections belong to $\mathcal{K}$. Extend each generated ideal from its affine chart to Y inside $\mathcal{K}$ by the first assertion, and take the ideal generated by these finitely many extended ideals. Its nonvanishing open is exactly W . $\square$

4.2. Modification into an open immersion. The following calculation is the affine input for modification.

Lemma 4.7. Let $A \in \mathbf{CMon}_0$ be a pointed commutative monoid, let $B \in \mathbf{CMon}_{0,A/}$ be of finite type, and let $t, s \in A$. Assume that t is cancellable in both A and B and that the given map induces an isomorphism

$$B[t^{-1}] \simeq A[(ts)^{-1}].$$

There are an integer $m \geq 1$, an element $b \in B$ with $sb = t^m$, and an A -algebra $A' \in \mathbf{CMon}_{0,A/}$ given by

$$A' = A[t^m/s] \subseteq A[s^{-1}]$$

which is the coordinate monoid of the s -chart of the blowup of $(t^m, s) \subseteq A$ and satisfies $A' \subseteq B \subseteq A'[t^{-1}]$. For suitable generators $b_1, \dots, b_r$ of B over A' and a common integer $N \geq 1$, write $b_j = c_j/t^N$ with $c_j \in A'$. Then B is the t^N -chart of the blowup of

$$(t^N, c_1, \dots, c_r) \subseteq A'.$$

Both ideals become the unit ideal after inverting t , and their pullbacks to B are principal ideals generated by cancellable elements.

Proof. The element s is invertible in $B[t^{-1}]$, so its inverse has the form b/t^m . Increasing m if necessary gives $m \geq 1$. Injectivity of $B \rightarrow B[t^{-1}]$ gives $sb = t^m$ in B . The same injectivity shows that s is cancellable in B . Consequently $(t^m, s)B = (s)$ is an invertible ideal. The universal property in [Lemma 4.4](#) gives a lift of $\mathrm{Spec}_0(B)$ to the s -chart, with t^m/s mapping to b . Multiplication by t remains injective in $A[s^{-1}]$, so

$$A' \hookrightarrow A'[t^{-1}] = A[(ts)^{-1}] = B[t^{-1}].$$

This identifies A' with a submonoid of B . The algebra B is finitely generated over A' , and clearing the finitely many powers of t gives the elements c_j . The t^N -chart of the second blowup is exactly

$$A'[c_1/t^N, \dots, c_r/t^N] = B.$$

Its ideal pulls back to $(t^N)B$, which is invertible. The assertions over $D(t)$ follow from the presence of t^m and t^N among the generators. $\square$

We now globalize this calculation, using the schematic images of [Section 2](#).

Proposition 4.8. Let $Y \in \mathbf{Sch}_0$ be a quasi-compact and quasi-separated pointed monoid scheme, and let $g : X \rightarrow Y$ be a separated morphism of finite type in $\mathbf{Sch}_0$. Let $U \in \mathbf{Sch}_{0,Y/}$ be a quasi-compact open subscheme such that $g^{-1}(U) \rightarrow U$ is an isomorphism. There is a finite sequence of blowups in quasi-coherent locally finitely generated ideals that are the unit ideal over U ,

$$p : Y' \rightarrow Y,$$

with $Y' \in \mathbf{Sch}_{0,Y/}$, such that the strict transform $X' \in \mathbf{Sch}_{0,Y'}$ has open-immersion structure morphism. The morphism p is proper and is an isomorphism over U .

Proof. We seek a diagram in Sch_0

$$\begin{array}{ccccc} U & \hookrightarrow & X' & \hookrightarrow & X \times_Y Y' \\ \parallel & & \downarrow & & \downarrow \\ U & \hookrightarrow & Y' & \xrightarrow{p} & Y \end{array}$$

in which X' is the schematic image of U in the fiber product and $X' \rightarrow Y'$ is an open immersion. We construct Y' locally, take a common refinement, and then glue the local inverses to $X' \rightarrow Y'$. By [Lemma 4.6](#), choose a locally finitely generated ideal whose nonvanishing open is U and first blow up this ideal. For the remainder of the construction replace Y by this blowup and X by its strict transform, retaining the same notation. By [Lemma 2.24](#), every subsequent strict transform relative to U is the strict transform for the composite modification of the original base. In each affine chart of the new base, the inverse image of U is $D(t)$ for a cancellable element t . Indeed, a blowup chart is a submonoid of a localization in which its exceptional generator is invertible. In an affine chart with monoid B_0 on the base change of X , the strict transform has coordinate monoid

$$\text{im}(B_0 \rightarrow B_0[t^{-1}]),$$

in which t is also cancellable. All these replacements preserve the original isomorphism over U .

Choose finitely many affine charts $\text{Spec}_0(B_i)$ of this strict transform, each lying over an affine chart $\text{Spec}_0(A_i)$ of the new base. Write $D(t_i)$ for the corresponding part of U . The morphism

$$\text{Spec}_0(B_i[t_i^{-1}]) \rightarrow \text{Spec}_0(A_i[t_i^{-1}])$$

is an affine open immersion. By [Lemma 2.12](#), every nonempty affine open in an affine monoid scheme is principal. Consequently, for some $s_i \in A_i$,

$$B_i[t_i^{-1}] \simeq A_i[(t_i s_i)^{-1}].$$

If $B_i[t_i^{-1}] = 0$, cancellability of t_i gives $B_i = 0$, so that chart may be omitted.

Apply [Lemma 4.7](#) to this pair of charts. The first ideal is defined on $\text{Spec}_0(A_i)$ and is the unit ideal on its intersection with U . Glue it to the unit ideal on U , and use [Lemma 4.6](#) to extend it over the base. Perform this blowup. Extend the second ideal from its specified blowup chart in the same way and perform the second blowup. Denote this two-step modification of the current base by $Y_i \rightarrow Y$. The chosen $\text{Spec}_0(B_i)$ lifts unchanged through both steps, since the pulled-back ideals are invertible, and becomes an open subscheme of Y_i . It is also its own strict transform: its graph in the base change is closed, and restriction to $D(t_i)$ is injective on its structure monoid.

These constructions for the finitely many original charts admit a common refinement by a finite sequence of blowups preserving U . To dominate one further blowup, blow up the pulled-back ideal; to dominate a tower, repeat this operation for its successive ideals. The universal property in [Lemma 4.4](#) supplies the required maps. Denote the resulting common refinement by $Y' \rightarrow Y$. The boundary of U is still locally defined by a cancellable element on Y' , since this property persists in blowup charts.

For each i , the inverse image in Y' of the open $\text{Spec}_0(B_i) \subseteq Y_i$ is an open $W_i \subseteq Y'$. Its graph in $\text{Spec}_0(B_i) \times_Y Y'$ is the pullback of the diagonal of $Y_i \rightarrow Y$. It is therefore closed, since $Y_i \rightarrow Y$ is separated. Restriction to $U \cap W_i$ is injective on the structure sheaf of W_i , by cancellability of the local boundary generator. Thus W_i is exactly the schematic image of $U \cap \text{Spec}_0(B_i)$ in that base change. The strict transform X' is therefore covered by open subschemes mapping isomorphically to the W_i .

It remains to check that these local open immersions give an open immersion globally. On $W_i \cap W_j$, the two induced sections of $X' \rightarrow Y'$ agree over U . Their equalizer is closed, by separatedness of $X' \rightarrow Y'$. On every affine open of $W_i \cap W_j$, restriction to its intersection with U is injective on the coordinate monoid. The quotient congruence defining the equalizer is therefore trivial. The two sections agree everywhere, so the local inverse maps glue.

Finally, each blowup is proper by [Lemma 4.4](#). Composition gives properness of p , and all its ideals restrict to the unit ideal over U . $\square$

This proposition supplies a direct monoid counterpart of the classical modification step [[The26](#), Tag 081S].

The following graph construction is the monoid version of [[The26](#), Tag 0F3Z].

Corollary 4.9. Let $S \in \text{Sch}_0$ be a quasi-compact and quasi-separated pointed monoid scheme, and let $X \in \text{Sch}_{0,/S}$ have separated structure morphism of finite type. Let $U \in \text{Sch}_{0,/X}$ be a quasi-compact open subscheme, and let $U \hookrightarrow Y$ be a compactification in $\text{Sch}_{0,/S}$ with $Y \rightarrow S$ proper. There are a proper modification $Y' \in \text{Sch}_{0,/Y}$ preserving U , an open subscheme $V \in \text{Sch}_{0,/Y'}$ containing U , and a proper morphism

$$V \rightarrow X$$

over S whose inverse image of U is U and whose restriction there is the identity.

Proof. The required extension is a diagram in $\text{Sch}_{0,/S}$

$$\begin{array}{ccccc} U & \hookrightarrow & V & \hookrightarrow & Y' \\ \parallel & & \downarrow & & \downarrow \\ U & \hookrightarrow & X & & Y. \end{array}$$

The left square is cartesian, the middle vertical morphism is proper, and the right vertical morphism is a modification preserving U . To construct it, take the schematic image Z of the graph map $U \rightarrow Y \times_S X$. The projection $Z \rightarrow Y$ is separated and of finite type and is an isomorphism over U . For the last assertion, restrict to $U \times_S X$, where the graph of $U \rightarrow X$ is closed by separatedness of $X \rightarrow S$. Apply [Proposition 4.8](#) to $Z \rightarrow Y$ and call its open strict transform $V \subseteq Y'$. The morphism $Z \rightarrow X$ is proper: it is the composite of a closed immersion into $Y \times_S X$ and a base change of $Y \rightarrow S$. The strict transform V is closed in $Z \times_Y Y'$, so $V \rightarrow Z$ is proper as well. Composition proves properness of $V \rightarrow X$. Restricting the original graph image to $Y \times_S U$ gives the graph of $U \rightarrow Y$, which is closed since $Y \rightarrow S$ is separated. All modifications preserve this graph over U , proving the assertion about its inverse image. $\square$

4.3. The boundary valuative criterion. For ordinary schemes, a separated morphism of finite type $f : Y \rightarrow S$ is proper if the valuative existence test holds for all diagrams whose generic-point map factors through a fixed quasi-compact dense open immersion $U \hookrightarrow Y$. Indeed, separatedness supplies uniqueness, and the dense valuative criterion [[The26](#), Tag 0894] gives universal closedness. One ingredient in that criterion is that, after passing to reduced schemes, a topologically dense open is schematically dense: a regular function vanishing on the open vanishes everywhere. Valuation rings and fields are reduced, so this passage does not change the test diagrams. The valuation-theoretic argument then recovers the full existence criterion from the restricted one.

The following example shows that, for pointed monoid schemes, testing the valuative criterion only on generic-point maps into a topologically dense open does not suffice to establish properness.

Example 4.10. Consider $A \in \text{CMon}_0$ and $T, Y, U \in \text{Sch}_0$ defined by

$$A = \mathbb{F}_1[a, b]/(ab \sim a), \quad T = \text{Spec}_0(A), \quad Y = D(b) \subseteq T, \quad U = D(a) \subseteq Y.$$

The prime ideals of A are $\{0\}$, (a) , and (a, b) . Thus $U = \{\{0\}\}$ is topologically dense in Y . The open immersion $Y \rightarrow T$ satisfies the valuative existence and uniqueness test whenever the generic-point map factors through U . In fact, for a valuation monoid $R \in \text{CMon}_0$, such a diagram sends a to a nonzero element of R . Cancellativity of R and $ab = a$ force the image of b to be 1, so the map to T

factors uniquely through $D(b)$. Nevertheless, $Y \rightarrow T$ is not proper. Take $R = \mathbb{F}_1[z] \in \mathbf{CMon}_0$ with pointed group completion $\mathbb{F}_1[z^{\pm 1}]$, and use the map

$$A \rightarrow R, \quad a \mapsto 0, \quad b \mapsto z.$$

The generic-point map factors through Y , but the map from $\mathrm{Spec}_0(R)$ does not, as z is not a unit. Thus testing only generic maps into an ordinary dense open loses necessary properness tests.

The difference from ordinary schemes is visible after realization over a field k :

$$Y_k = \mathrm{Spec}(k[a, b, b^{-1}]/(a(b-1))), \quad U_k = D(a) \subseteq Y_k.$$

Here Y_k has two irreducible components, $V(a)$ and $V(b-1)$, and the closure of U_k is only $V(b-1)$. Thus U_k is not dense in Y_k , although U is dense in the ordinary prime-ideal space of Y . The ordinary dense valuative criterion therefore does not apply to this realization.

We repair the density issue by arranging a boundary which is locally defined by a cancellable element. This is stronger than ordinary density and can be obtained by blowing up an ideal defining the complement of the prescribed open.

Lemma 4.11. Let $A \in \mathbf{CMon}_0$ be a pointed commutative monoid, and let $t \in A$ be cancellable. Every point of $\mathrm{Cong}(A)$ is a specialization of a point in $\mathrm{Cong}(A[t^{-1}])$. Consequently, if $Y \in \mathbf{Sch}_0$ and an open subscheme $U \in \mathbf{Sch}_{0,/Y}$ is locally of this form, then $\mathrm{Cong}(U)$ is dense in $\mathrm{Cong}(Y)$. every proper open immersion $V \rightarrow Y$ in $\mathbf{Sch}_0$ whose image contains U is an isomorphism.

Proof. Let $\mathfrak{p}$ be a prime congruence of A , let I be its null ideal, and set

$$C = A[(A \setminus I)^{-1}], \quad D = C[t^{-1}].$$

The map $C \rightarrow D$ is injective, as multiplication by t remains injective after localization. Let J be the inverse image in A of the nonunits of D . Then J is a prime ideal, $J \subseteq I$, and $t \notin J$. Consider the prime congruence $\mathfrak{q}$ given by the kernel of

$$A \rightarrow (A \setminus J)^{\mathrm{gp}} \cup \{0\},$$

where all elements of J map to zero. We claim that $\mathfrak{q} \subseteq \mathfrak{p}$. Pairs of elements of J already belong to $\mathfrak{p}$. If $x, y \notin J$ have the same group-completion image, then $cx = cy$ in A for some $c \notin J$. The image of c is invertible in D , so $x = y$ in D and hence in C . The map from A to the pointed fraction group of $A/\mathfrak{p}$ extends to C , proving $x \sim_{\mathfrak{p}} y$. This proves the claim. Thus $\mathfrak{q}$ generalizes $\mathfrak{p}$ and belongs to the indicated open.

The assertion for schemes follows on affine charts. For the last assertion, the image on congruence spaces of a proper open immersion is closed and contains this dense open, hence is the entire congruence space. Every ordinary monoid prime is the null ideal of a prime congruence, so the underlying open subset of Y is all of Y . $\square$

The following restricted test retains arbitrary maps of the generic point into the chosen open.

Lemma 4.12. Let $S \in \mathbf{Sch}_0$ be a quasi-compact and quasi-separated pointed monoid scheme, and let $f : Y \rightarrow S$ be a separated morphism of finite type in $\mathbf{Sch}_0$. Let $U \in \mathbf{Sch}_{0,/Y}$ be a quasi-compact open subscheme such that every point of Y has an affine neighborhood $\mathrm{Spec}_0(A)$, with $A \in \mathbf{CMon}_0$, on which $U = D(t)$ for a cancellable element $t \in A$. Then f is proper if and only if every valuation-monoid diagram

$$\begin{array}{ccccc} \mathrm{Spec}_0(K) & \longrightarrow & U & \hookrightarrow & Y \\ \downarrow & & \nearrow & & \downarrow f \\ \mathrm{Spec}_0(R) & \longrightarrow & & & S \end{array}$$

with $R \in \mathbf{CMon}_0$ a valuation monoid and $K \in \mathbf{CMon}_{0,R/}$ its pointed fraction group admits a lift to Y .

Proof. Necessity is [Theorem 3.6](#). For sufficiency, separatedness makes all such lifts unique. Fix a field k and form the ordinary monoid-algebra realizations $U_k \subseteq Y_k \rightarrow S_k$. On each specified affine chart the map

$$k[A] \rightarrow k[A[t^{-1}]] = k[A][t^{-1}]$$

is injective: multiplication by t is injective on the monomial basis. Consequently U_k is schematically dense, and in particular topologically dense, in Y_k . Both $U_k \rightarrow Y_k$ and $Y_k \rightarrow S_k$ are quasi-compact.

For every valuation ring over k , its multiplicative pointed monoid is a valuation monoid and its pointed fraction group is the multiplicative pointed monoid of its fraction field. The local monoid-algebra adjunction [[CHWW15](#), Theorem 5.2, equation (5.2a)] transfers the assumed lifting property to every ordinary valuation-ring diagram with generic point in U_k . The ordinary dense valuative criterion [[The26](#), Tag 0894] shows that $Y_k \rightarrow S_k$ is universally closed. It is separated and of finite type by [Proposition 2.17](#), hence proper. By [Theorem 3.7](#), every valuation-monoid diagram for $Y \rightarrow S$ therefore admits a unique lift. Applying [Theorem 3.6](#) proves the assertion. This argument uses realization to verify properness of a constructed monoid scheme; it requires no descent of an ordinary compactification. $\square$

4.4. Gluing compactifications. We record the strict-transform calculation used to compare different boundary modifications.

Lemma 4.13. Let $g : Y \rightarrow Z$ be a morphism in Sch_0 with Y, Z quasi-compact and quasi-separated, and let $U \in \text{Sch}_{0,/Y}$ be a quasi-compact schematically dense open subscheme. Let $\mathcal{I} \subseteq \mathcal{O}_Z$ be a quasi-coherent locally finitely generated ideal sheaf of pointed $\mathcal{O}_Z$ -sets whose extension to U is the unit ideal. The schematic image of the canonical lift of U in

$$Y \times_Z \text{Bl}_{\mathcal{I}}(Z)$$

is canonically isomorphic in $\text{Sch}_{0,/Y}$ to $\text{Bl}_{\mathcal{I}\mathcal{O}_Y}(Y)$. In particular, every two finite towers of blowups of Y in such ideals admit a common refinement in $\text{Sch}_{0,/Y}$ which is a finite tower of such blowups over each of them.

Proof. Work with affine charts $\text{Spec}_0(B) \rightarrow \text{Spec}_0(A)$, let I be the ideal on the second chart, and put $J = IB$. For a generator $a \in I$ with image $b \in B$, the two relevant charts have coordinate monoids

$$B \amalg_A A[I/a] \rightarrow B[J/b] \subseteq B[b^{-1}].$$

The map is surjective, as the fractions j/b generate its target over B . Schematic density, followed by principal localization, gives an injection

$$B[b^{-1}] \hookrightarrow \Gamma(U \cap \text{Spec}_0(B) \cap D(b), \mathcal{O}_U).$$

The intersection of the lifted U with the chosen blowup chart is precisely this principal open: the ideal is the unit ideal on U . It follows that the kernel congruence of restriction from the base-change chart is exactly the kernel of the indicated surjection. These identifications glue and prove the first assertion. They also show that the lifted U remains schematically dense after every blowup.

For two towers $Y_1 \rightarrow Y$ and $Y_2 \rightarrow Y$, take the schematic image of U in $Y_1 \times_Y Y_2$. Applying the first assertion successively to the ideals of the second tower identifies its projection to Y_1 with a tower of blowups in the pulled-back ideals. Here [Lemma 2.24](#) identifies the successive strict transforms with the schematic image of U in the final fiber product. Interchanging the towers gives the same assertion over Y_2 . All these ideals are the unit ideal on U . $\square$

The next lemma extends an isomorphism near a prescribed closed subset to its boundary, following [[The26](#), Tag 0F3W].

Lemma 4.14. Let $Y \in \text{Sch}_0$ be a quasi-compact and quasi-separated pointed monoid scheme, and let $f : Z \rightarrow Y$ be a proper morphism in Sch_0 . Let $V \in \text{Sch}_{0,/Y}$ be a quasi-compact open subscheme, and let $T \subseteq |V|$ be a closed subset of its underlying topological space. Assume that f is an isomorphism over an open neighborhood of T in V . There is a finite tower of blowups $Y' \rightarrow Y$ in Sch_0 , with centers given by locally finitely generated quasi-coherent ideal sheaves that are the unit ideal on V , such that the strict transform of f is an isomorphism over an open neighborhood of the ordinary closure of T in Y' . The strict transform here is the schematic image of $f^{-1}(V)$ in $Z \times_Y Y'$.

Proof. Choose a quasi-compact open $G \subseteq V$ containing T over which f is an isomorphism. This is possible as T is a closed subset of the quasi-compact space V . Put $T' = V \setminus G$. Choose a quasi-compact open of $V \setminus T$ containing T' , and let T_c be its complement in V . Then $T \subseteq T_c \subseteq G$, and T_c, T' are disjoint closed subsets with quasi-compact complements in V . By Lemma 4.6, they are defined by locally finitely generated ideals on V which extend to Y . Blow up the union of these extensions and use the divided ideals in Lemma 4.5. Their vanishing sets are disjoint closed extensions of T_c and T' . Replace Y by this blowup and Z by the indicated strict transform.

Let Y_0 be the complement of the closed extension of T' . This is a quasi-compact open, contains the closure of T , and satisfies $Y_0 \cap V = G$. Apply Proposition 4.8 to $Z \times_Y Y_0 \rightarrow Y_0$ with fixed open G . Its new strict transform is both proper and an open subscheme of the modified Y_0 . The construction begins by blowing up the complement of G , so G is locally the complement of a cancellable principal divisor on this modified base. By Lemma 4.11, the proper open immersion is an isomorphism.

Extend each ideal used on Y_0 to the current whole base, after gluing it to the unit ideal on V . This is possible by Lemma 4.6, since its restriction to $Y_0 \cap V = G$ is the unit ideal. The resulting global tower preserves V and restricts to the constructed tower over Y_0 . Open restriction of schematic images identifies its strict transform over that open with the isomorphism just obtained. The inverse image of the closed extension of T_c contains the closure of T and lies inside this open. This proves the assertion. $\square$

One further classical shortcut must be replaced. A proper locally closed immersion of monoid schemes need not be closed. For example, the two copies of $\text{Spec}_0(\mathbb{F}_1)$ mapping to $x = 1$ and $y = 1$ in $\text{Spec}_0(\mathbb{F}_1[x, y]/(xy \sim 0))$ form such an immersion; its image on congruence spaces is closed, but the morphism is not affine [LR24, Remark 7.5]. It is proper by the valuative criterion, since either specified point has a unique extension through every valuation monoid. For the diagonal, however, a closed image on congruence spaces does characterize separatedness, provided the morphism is quasi-separated [LR24, Theorem 7.6].

Proposition 4.15. Let $S \in \text{Sch}_0$ be a quasi-compact and quasi-separated pointed monoid scheme, and let $Y_1, Y_2 \in \text{Sch}_{0,/S}$ be separated and of finite type. Let $U \in \text{Sch}_{0,/S}$ be equipped with open immersions into both Y_i , and assume that U is quasi-compact. There are finite towers of blowups $Y'_i \rightarrow Y_i$ in $\text{Sch}_{0,/S}$, with centers given by locally finitely generated quasi-coherent ideal sheaves that are the unit ideal on U , and open immersions

$$Y'_1 \hookrightarrow W \hookleftarrow Y'_2$$

in $\text{Sch}_{0,/S}$ which agree on U , with $W \in \text{Sch}_{0,/S}$ separated and of finite type. Moreover, every point of W has an affine neighborhood $\text{Spec}_0(A)$, with $A \in \text{CMon}_0$, on which $U = D(t)$ for a cancellable element $t \in A$.

Proof. We construct an object $H \in \text{Sch}_{0,/S}$ with compatible open immersions into modifications Y'_1 and Y'_2 of the given schemes. Their gluing is the pushout

$$W = Y'_1 \amalg_H Y'_2$$

in $\text{Sch}_{0,/S}$. The additional requirement is that $H \rightarrow Y'_1 \times_S Y'_2$ be proper; this will imply separatedness of W through the congruence-space criterion for its diagonal. Let Z be the schematic image of the

common graph $U \rightarrow Y_1 \times_S Y_2$. The open U is schematically dense in Z , and each projection $Z \rightarrow Y_i$ is an isomorphism over U . Indeed, over $U \times_S Y_i$ the graph is closed by separatedness of $Y_i \rightarrow S$. Apply [Proposition 4.8](#) to each projection. It gives a tower $Y_i^{(i)} \rightarrow Y_i$ and an open strict transform $Z_i \subseteq Y_i^{(i)}$. By [Lemma 4.13](#), each $Z_i \rightarrow Z$ is a finite tower of blowups preserving U . The same lemma gives a common refinement $H \rightarrow Z_i$ which is a finite tower of such blowups for each i .

Extend the successive ideals of $H \rightarrow Z_i$ from the open Z_i to $Y_i^{(i)}$ using [Lemma 4.6](#). Blowing up these extensions gives $Y_i' \rightarrow Y_i^{(i)}$ with H open in Y_i' . All modifications preserve U . The first modification on each Y_i makes its boundary locally cancellable and principal, and this property persists in further blowup charts.

The morphism $H \rightarrow Y_1' \times_S Y_2'$ is proper. Its composite with $Y_1' \times_S Y_2' \rightarrow Y_1 \times_S Y_2$ factors through the proper map $H \rightarrow Z$ and the closed immersion of Z . Since the morphism between these products is separated, the closed-graph factorization proves properness of the map to the modified product.

Form the indicated pushout $W = Y_1' \amalg_H Y_2'$ in $\text{Sch}_{0,/S}$. It is of finite type over S and quasi-separated, as the two pieces and their intersection are quasi-compact and quasi-separated. On each product $Y_i' \times_S Y_i'$, the diagonal of $W \rightarrow S$ has closed image on congruence spaces by separatedness of $Y_i' \rightarrow S$. On the two mixed products its restriction is $H \rightarrow Y_1' \times_S Y_2'$ or its transpose, whose image on congruence spaces is closed by properness. These four products cover $W \times_S W$, so the diagonal has closed image on its congruence space. By [\[LR24, Theorem 7.6\]](#), $W \rightarrow S$ is separated. The local description of the boundary holds on the two open pieces and hence on W . $\square$

We now adapt the binary gluing argument of [\[The26, Tag 0F40\]](#).

Proposition 4.16. Let $S \in \text{Sch}_0$ be a quasi-compact and quasi-separated pointed monoid scheme, and let $X \in \text{Sch}_{0,/S}$ have separated structure morphism of finite type. Suppose $X = X_1 \cup X_2$, where $X_1, X_2 \in \text{Sch}_{0,/X}$ are quasi-compact open subschemes admitting proper compactifications in $\text{Sch}_{0,/S}$. Then X admits an open immersion in $\text{Sch}_{0,/S}$ into an object $\overline{X} \in \text{Sch}_{0,/S}$ with proper structure morphism.

Proof. Choose compactifications $X_i \subseteq Y_i$. By [Corollary 4.9](#), after modifications preserving X_i there are quasi-compact opens $X_i \subseteq V_i \subseteq Y_i$ and proper maps

$$\psi_i : V_i \rightarrow X, \quad \psi_i^{-1}(X_i) = X_i,$$

which are the identity on X_i . For $\{i, j\} = \{1, 2\}$, set

$$Z_i = X \setminus X_j, \quad Z_{i,i} = \psi_i^{-1}(Z_i), \quad Z_{i,j} = \psi_i^{-1}(Z_j).$$

The last two subsets are disjoint and closed in V_i and have quasi-compact complements there. Extend ideals defining them to Y_i and apply [Lemma 4.5](#). After these modifications preserving V_i , their ordinary closures

$$C_{i,i} = \overline{Z_{i,i}}, \quad C_{i,j} = \overline{Z_{i,j}}$$

in Y_i are disjoint. Disjointness persists after every further modification preserving V_i , since the new closures lie in the inverse images of the old ones.

Put $V_{12} = V_1 \times_X V_2$, and let Y_{12} be its schematic image in $Y_1 \times_S Y_2$. The morphism $V_{12} \rightarrow V_1 \times_S V_2$ is a closed immersion, being a base change of the diagonal of $X \rightarrow S$. Since $V_1 \times_S V_2$ is open in $Y_1 \times_S Y_2$, open restriction of schematic images identifies the restriction of Y_{12} to this open with V_{12} . The projections $p_i : Y_{12} \rightarrow Y_i$ are proper, and V_{12} is a schematically dense open of Y_{12} . For

$$G_1 = \psi_1^{-1}(X_2) \subseteq V_1,$$

the inverse image of G_1 in V_{12} is G_1 , and its map into $G_1 \times_S Y_2$ is the graph of the morphism $G_1 \rightarrow Y_2 \subseteq Y_2$. This graph is closed in $G_1 \times_S Y_2$, by separatedness of $Y_2 \rightarrow S$. Open restriction of the schematic image therefore shows that p_1 is an isomorphism over G_1 .

Since $Z_{1,2} \subseteq G_1$, apply [Lemma 4.14](#) with fixed open V_1 and closed subset $Z_{1,2}$. Replace Y_1 by the resulting modification and Y_{12} by its strict transform. To compare the defining opens, write $E = p_1^{-1}(V_1)$ on the original Y_{12} . The open $V_{12} \subseteq E$ is schematically dense, by restriction of its schematic density in Y_{12} . The modification preserves V_1 , so its base change preserves E . The strict transform specified in [Lemma 4.14](#) is the schematic image of this unchanged E . By [Lemma 2.24](#), it is also the schematic image of V_{12} in the base change of Y_{12} , and hence in the new product, since that base change is a closed subscheme of the new product. We may thus assume that p_1 is an isomorphism near $C_{1,2}$.

It follows that

$$p_1^{-1}(C_{1,2}) \cap p_2^{-1}(C_{2,1}) = \emptyset.$$

Indeed, $p_1^{-1}(Z_{1,2})$ is dense in $p_1^{-1}(C_{1,2})$, since p_1 is an isomorphism near the latter set. Its image under p_2 lies in $Z_{2,2}$. Continuity therefore puts the image of $p_1^{-1}(C_{1,2})$ inside $C_{2,2}$, which is disjoint from $C_{2,1}$.

Choose quasi-compact opens

$$O_i \subseteq Y_i \setminus C_{i,j}$$

which contain $X_i \cup C_{i,i}$ and satisfy

$$Y_{12} = p_1^{-1}(O_1) \cup p_2^{-1}(O_2).$$

Here no Noetherian hypothesis is needed. The two full complements have inverse images covering the quasi-compact space Y_{12} by the preceding disjointness statement. Refine this cover by inverse images of affine opens contained in the corresponding complements and choose finitely many. Enlarge their finite unions by finitely many affine opens covering $X_i \cup C_{i,i}$ inside the same complement. Such a finite enlargement exists since X_i is quasi-compact and $C_{i,i}$ is closed in the quasi-compact space Y_i .

Form the pushouts in $\text{Sch}_{0,/S}$

$$W_i = X \coprod_{X_i} O_i.$$

They are of finite type over S and quasi-separated. We verify their separatedness valuatively. Only the case of two extensions through different pieces needs attention. Write them as $a : \text{Spec}_0(R) \rightarrow X$ and $b : \text{Spec}_0(R) \rightarrow O_i$, with common generic-point map in X_i . Properness of ψ_i lifts a to $c : \text{Spec}_0(R) \rightarrow V_i$ with the prescribed generic point. The maps c and b agree as maps to Y_i , since $Y_i \rightarrow S$ is separated. If the closed point of a lay in $Z_j = X \setminus X_i$, the closed point of c would lie in $Z_{i,j} \subseteq C_{i,j}$, contradicting $b(\text{Spec}_0(R)) \subseteq O_i$. Thus a factors through X_i . The equality $\psi_i^{-1}(X_i) = X_i$ shows that c and b also factor through this common open. The two extensions agree in W_i , and [\[LR24, Theorem 7.7\]](#) proves separatedness.

We next show that every valuation diagram over S with generic-point map into X extends to at least one W_i . Write K for the pointed fraction group of the valuation monoid R . If the image of $\text{Spec}_0(K) \rightarrow X$ lies in Z_i , use properness of $Y_i \rightarrow S$ to extend it to $\text{Spec}_0(R) \rightarrow Y_i$. The image of its closed point lies in the ordinary closure of $Z_{i,i}$, hence in $C_{i,i} \subseteq O_i$. The extension therefore factors through $O_i \subseteq W_i$.

If the generic-point image lies in $X_1 \cap X_2$, properness gives extensions to both Y_1 and Y_2 . Their pair factors through Y_{12} . To see this, its generic-point map factors through V_{12} , and every defining congruence of the closed immersion $Y_{12} \rightarrow Y_1 \times_S Y_2$ which vanishes in K already vanishes in $R \subseteq K$. The covering $Y_{12} = p_1^{-1}(O_1) \cup p_2^{-1}(O_2)$ then puts the closed point of at least one extension in O_i . This gives the desired extension to W_i . These cases exhaust all generic-point maps into X .

Apply [Proposition 4.15](#) to W_1, W_2 with common open X . We obtain proper modifications $W'_i \rightarrow W_i$ preserving X and a separated finite-type scheme $\overline{X}$ obtained by gluing the modified pieces. Each extension to W_i lifts to W'_i by properness, since its generic point is already in the unchanged open X . Hence every valuation diagram with generic-point map in X has a unique lift to $\overline{X}$. The gluing

proposition makes X locally the complement of a cancellable principal divisor in $\overline{X}$. By Lemma 4.12, the morphism $\overline{X} \rightarrow S$ is proper. $\square$

The theorem now follows by finite induction.

Theorem 4.17. Let $S \in \text{Sch}_0$ be a quasi-compact and quasi-separated pointed monoid scheme, and let $f : X \rightarrow S$ be a separated morphism of finite type in Sch_0 . There is a factorization

$$X \xrightarrow{j} \overline{X} \xrightarrow{\bar{f}} S$$

with $\overline{X} \in \text{Sch}_{0/S}$, where j is a quasi-compact open immersion in $\text{Sch}_{0/S}$ and $\bar{f}$ is proper. The compactification can be chosen so that every point of $\overline{X}$ has an affine neighborhood $\text{Spec}_0(A)$, with $A \in \text{CMon}_0$, such that $X \cap \text{Spec}_0(A) = D(t)$ for a cancellable element $t \in A$.

Proof. The assertion is immediate for empty X . Otherwise choose a finite affine open cover $X = \bigcup_{i=1}^n U_i$, with each U_i mapping into an affine open $S_i \subseteq S$. By Proposition 4.3, each U_i has a proper compactification over S . Every finite union of the U_i is a quasi-compact open of X , hence is separated and of finite type over S . Induction using Proposition 4.16 yields a compactification of X . Finally, blow up a locally finitely generated ideal whose nonvanishing open is X , supplied by Lemma 4.6. This modification is proper, preserves X , and gives the asserted local description of its boundary. Finally, X is quasi-compact because it is of finite type over the quasi-compact base S , and $\overline{X}$ is quasi-separated because it is separated over the quasi-separated base S . Thus the open immersion j is quasi-compact, as required in the definition of compactification. $\square$

We conclude by noticing that Nagata compactification can be deduced easily in the toric case.

Remark 4.18. For a separated, connected, normal, torsion-free monoid scheme X of finite type over $\mathbb{F}_1$, compactification follows from completion of fans. Indeed, [CHWW15, Theorem 4.4] identifies X with the monoid scheme X_Σ associated to a finite rational fan Σ in a lattice N . There is a complete rational fan $\overline{\Sigma}$ containing Σ as a subfan; see Ewald-Ishida [EI06] and Rohrer [Roh11] for constructive proofs. The resulting open immersion

$$X_\Sigma \hookrightarrow X_{\overline{\Sigma}}$$

is a compactification: completeness of $\overline{\Sigma}$ gives properness by [CHWW15, Corollary 8.16], whose valuative formulation agrees with the notion used here by Theorem 3.6.

For a morphism $X \rightarrow S$ between such toric monoid schemes, choose an absolute compactification $X \hookrightarrow P$ as above and take the schematic image of its graph in $P \times_{\mathbb{F}_1} S$. This image is proper over S , and its restriction to $X \times_{\mathbb{F}_1} S$ is the graph, since S is separated over $\mathbb{F}_1$. It therefore gives a compactification of X over S . The theorem proved here extends this existence result to arbitrary pointed monoid schemes over quasi-compact and quasi-separated bases. Such schemes need not admit a description by fans, and compactifying a normalization alone does not supply a compactification of the original scheme.

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